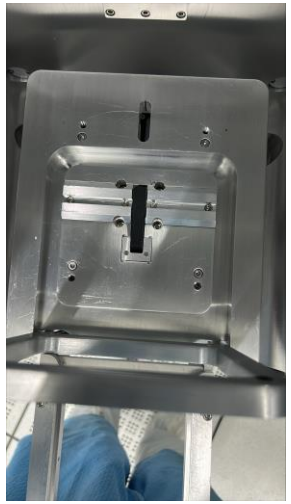
Tool questions with answers

- Vintage/tool age?
  Tool 1: August 2015. Tool 2: Combination of 3 chamber to one TM, Feb-2017 (PM1), Jun-2015 (PM2), Aug-2015 (PM3), Mar 2014 (TM)
- How many process chambers?
  2 systems. One with 2 chambers and one with 3 chambers.
- What type of process chambers are installed?
  All are ICP chambers
- What gas config is running through the tool?
  8 gases. CF4, O2_a, He, Ar, O2_b, CHF3, N2, H2 (please refer to next slide and slide 28)
- What kind of matching networks are installed?
  RF matching network, Seren. (please refer to slide 21)
- What kind of turbo pumps are installed?
  BOC Edwards, STP-A1303C turbomolecular pump (please refer to slide 25)
- What kind of Robot is installed in the transfer module?
  Brooks MagnaTran 7 robot (refer to slide 24)
- What kind of loaders are present on the Front-End module?
  6” Cassette loader.
- Is the control system based on Cortex ?
  Cortex control system
- Does it have an electrostatic chuck or a mechanical wafer clamp?
  Electrostatic chuck
- Does it include the mechanical pumps?
  Edwards IH80 for chambers and IGX100 for TM.
- Is there hardware for 4” and 6"?
  6”
- Serial number?
  Tool 1: 101020039. Tool 2: 101020215, 101020045, 101020040, 101019928, Refer to slides 4 and 12

Photos Needed:

Front-End module



Cassette loader Hine elevator

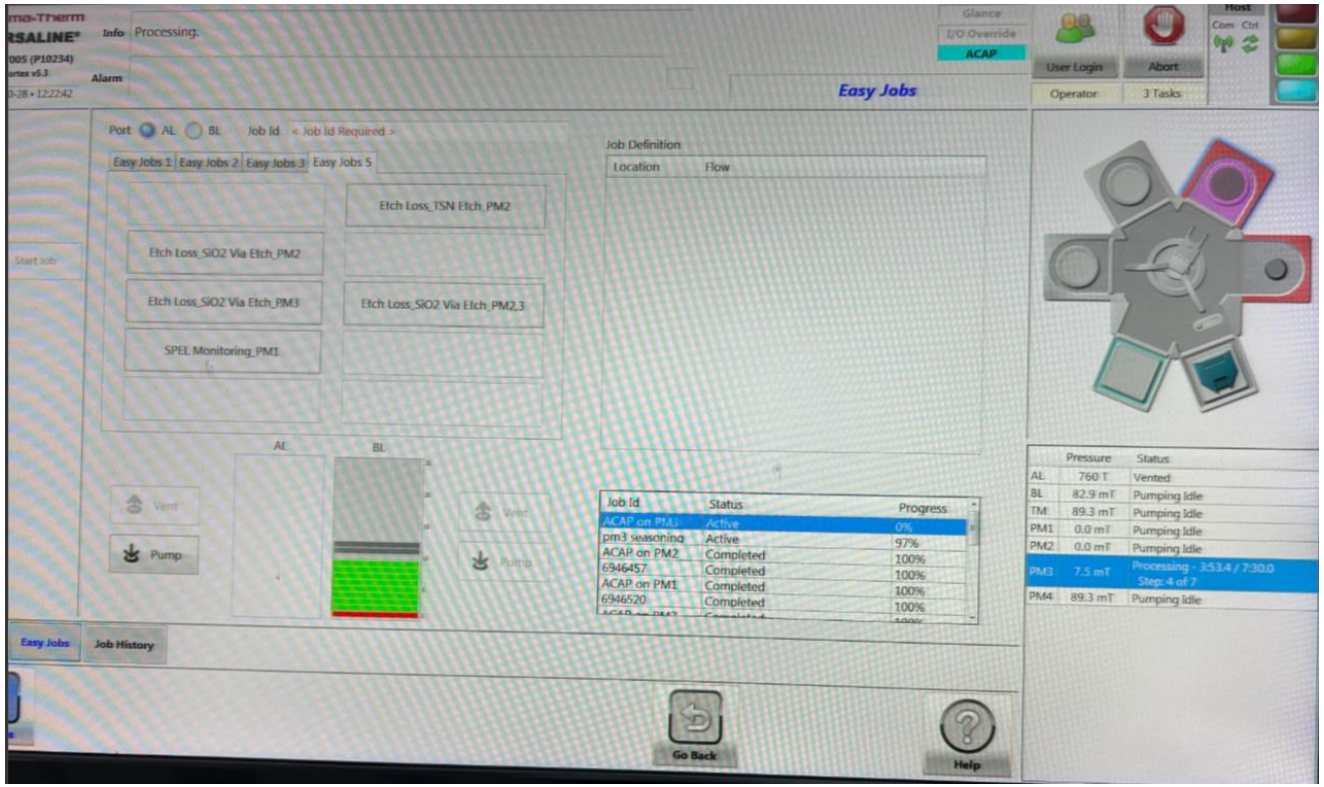
Tool 1

2 loadlocks

Tool 2

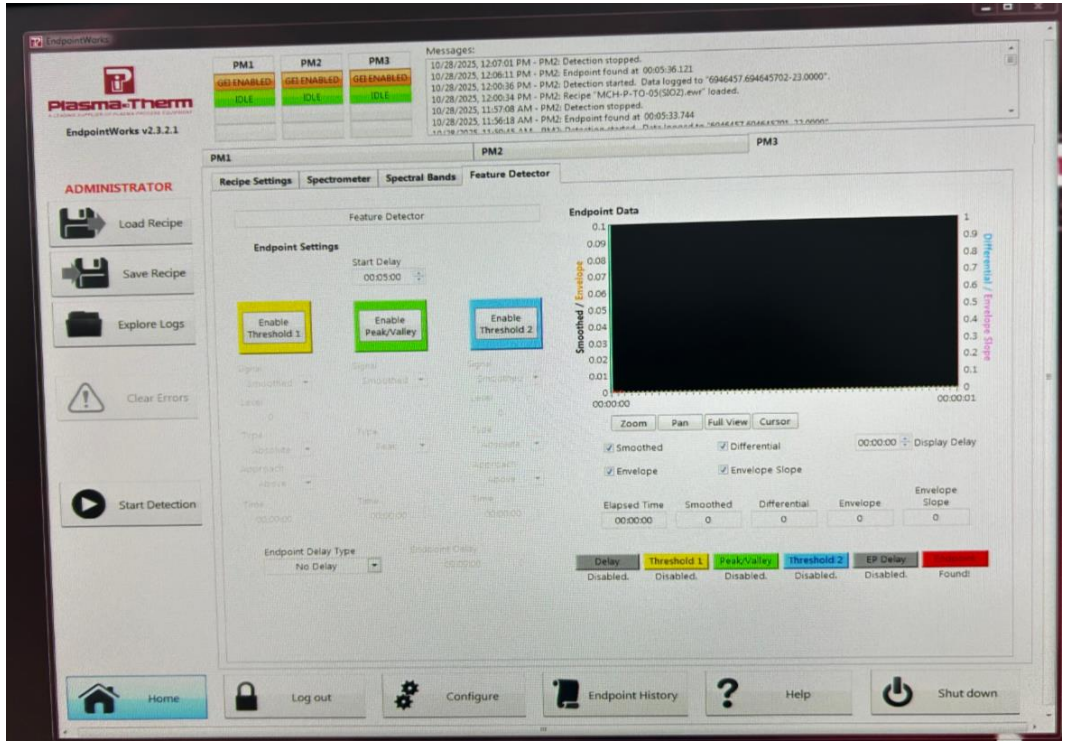
Photos Needed:

Control Station, including SW



Cortex control station

CTC PC and UPS



Endpoint Work

Photos Needed:

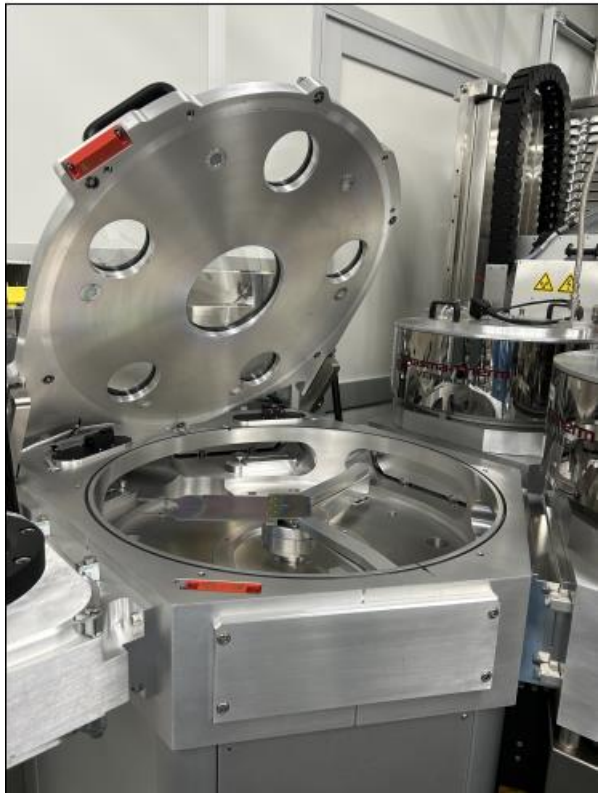
Photo of the Transfer Module

Tool with 3 chambers with transfer module

Tool with 2 chambers with transfer module

Photos Needed:

Photo of the robot underneath the TM



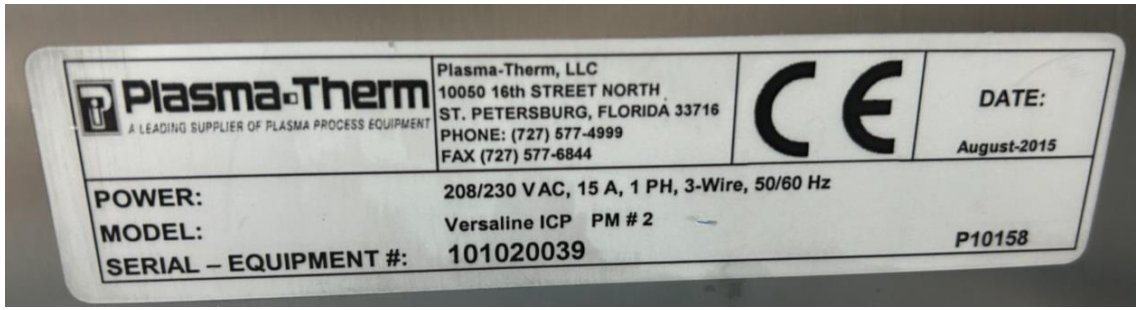
Robot arm

Brooks MagnaTran 7 robot

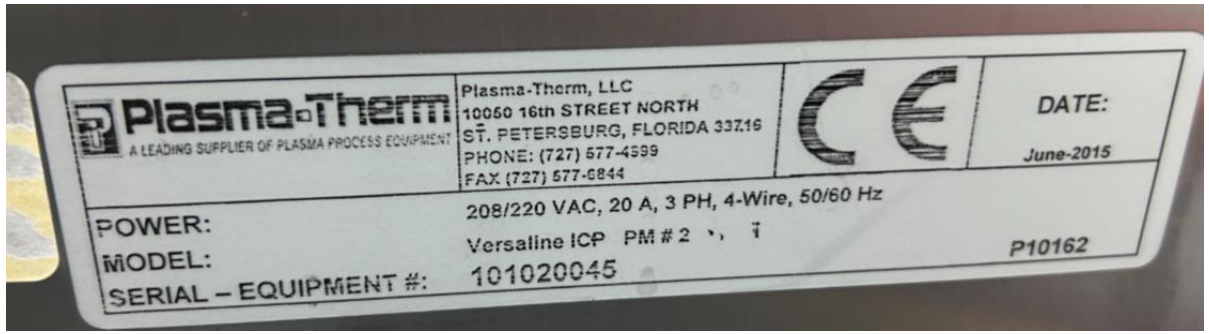
Photos Needed:

Serial Tag on the TM

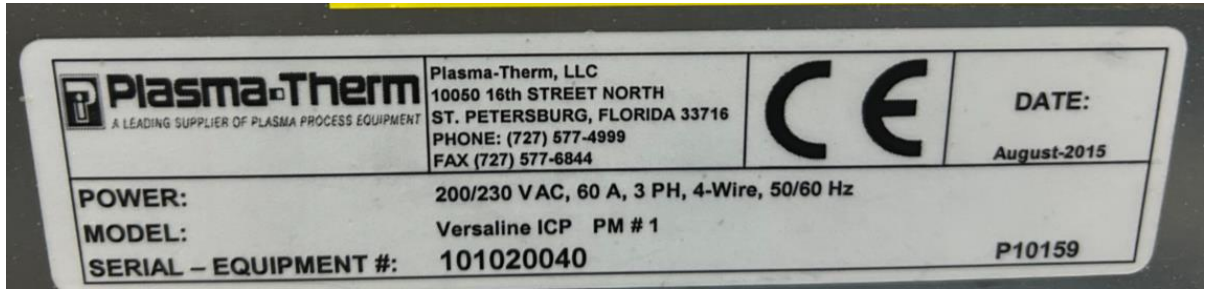
TM



Serial tag for tool 1 (2 chambers system)



PM1



PM2

Serial tag for tool 2 (3 chambers system)
Note that for this system the chambers are combines from different systems

PM3

Photos Needed:

Photo of each Process Module

Process module sideview

Photos Needed:

Power distribution unit

Power distribution units for the 3 process module and TM

Power distribution Unit for the 2 process modules and TM

Photos Needed:

Turbo pump

Turbo pump

Photos Needed:

Matching networks

ENI RF generator for ICP power

Elite RF generator for bias power

Elite RF generator for bias power (type 2)

Photos Needed:

Matching networks

Seren ICP matcher

Seren bias matcher

Photos Needed:

Matching networks

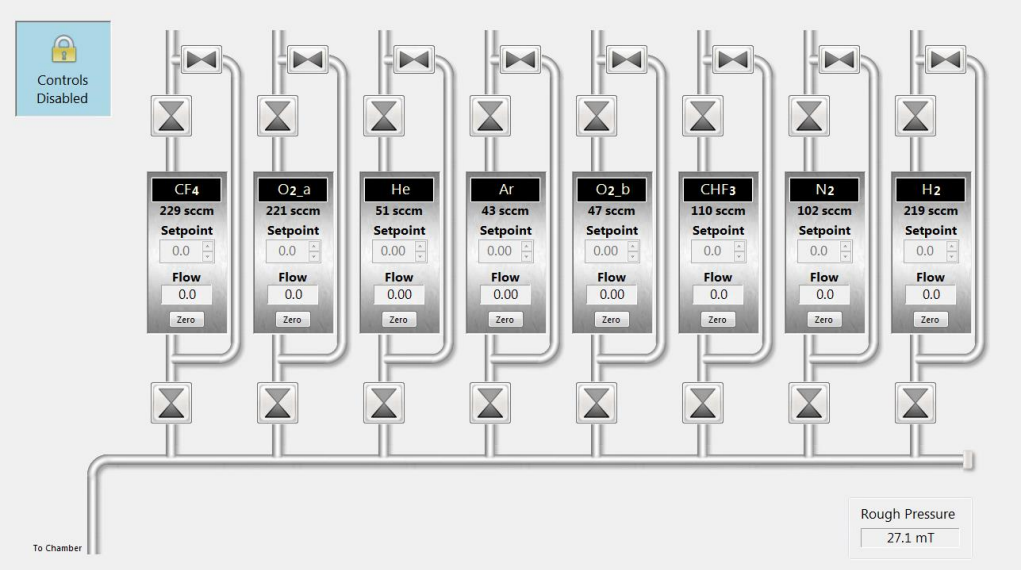
Turbo pump controller

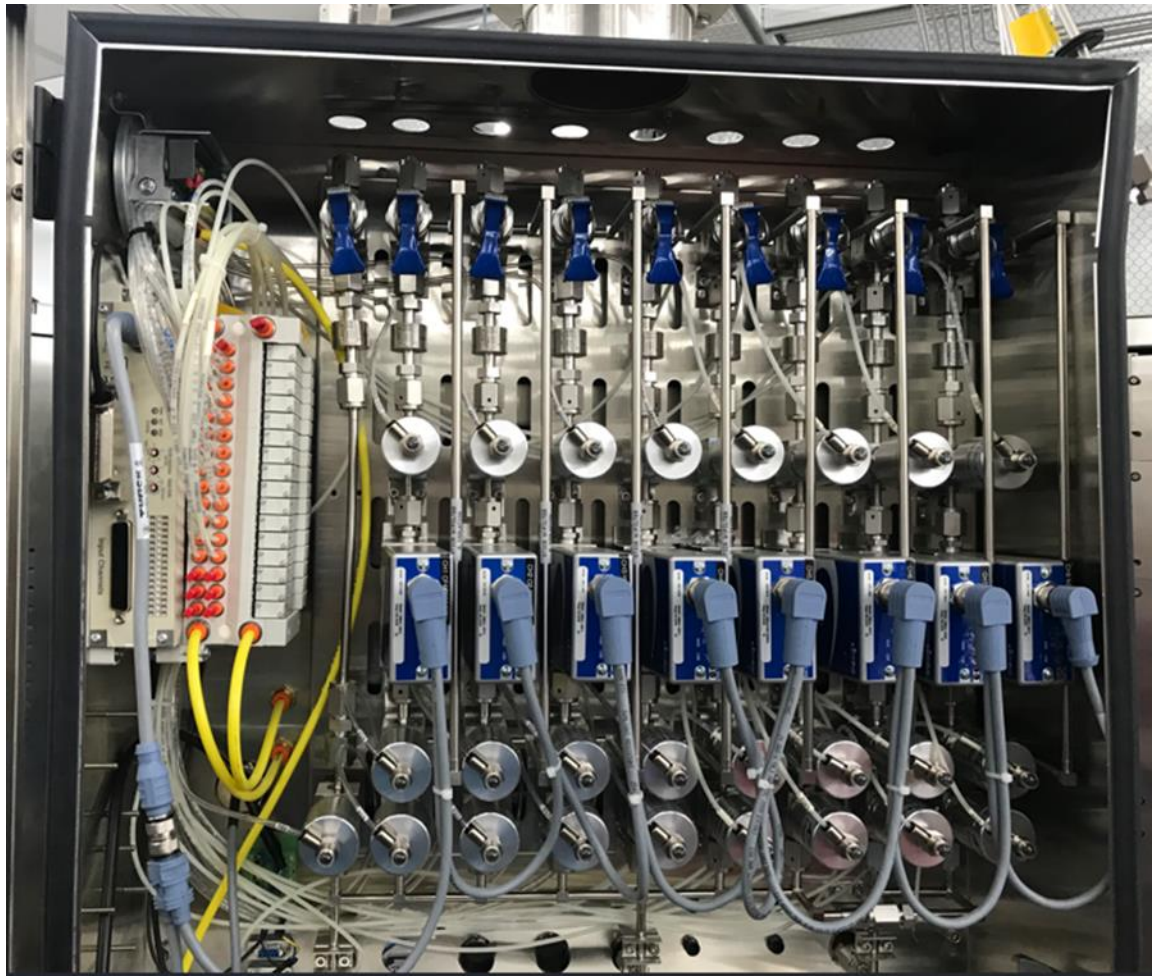
Matching networks
for ICP and bias

ESC power supply

Photos Needed:

Gas box for each PM





Photos Needed:

Control PC in the rear of the tool

Rear control PC

Photos Needed:

Photos of the pumps included with the tool

Rough pumps

Photos Needed:

Photos of the chillers included with the tool

Chillers

Photos Needed:

If there is a spare change kit,
a photo of that

There is No spare kit